

Spatial aggregation alters population-weighted estimates of InSAR observation support

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Abstract

Population-weighted assessments of interferometric synthetic aperture radar (InSAR) observation support depend on how population is allocated within reporting units. Spatial aggregation can change these assessments even when the observations are unchanged. We isolate this effect by comparing allocation rules under a fixed geographic domain, support field and population total. In the Chao Phraya footprint, uniform allocation within 1-km cells increases the mean support deficit by 57.7–57.9% relative to native population-model allocation, equivalent to 7.75–7.78 percentage points of the domain total. Within-cell covariance between population and support explains this difference, which persists after excluding boundary cells. Crossed population-model and interferogram-sampling comparisons preserve its direction but change its magnitude. Shifts as a share of total population are smaller for the final processing mask and in the California cases. Controlled missingness shows that small errors where estimates remain available can coexist with substantial unestimated population allocation. The primary deficit averages unsupported model allocation over area and interferogram pairs; it does not identify subsidence exposure. Population-weighted reporting should therefore preserve the spatial association between population and observation support and report sensitivity to allocation and sampling.

Keywords: InSAR, population allocation, observation support, coherence, spatial scale, reproducibility

1 Introduction

Satellite radar missions such as Sentinel-1 provide repeated observations for deformation mapping [1]. Classical reviews describe its interferometric basis and geophysical applications [2–5]. Persistent-scatterer, small-baseline and later multi-temporal methods extend monitoring across point-like and distributed scatterers [6–9]. Public processing chains and regional products have broadened access to urban and deltaic subsidence measurements [10–12]. For downstream screening, deformation layers must be translated into population summaries without losing the support limitations recorded in the product.

This translation requires spatial accounting because InSAR does not observe all surfaces equally. Temporal decorrelation and surface change can reduce measurement validity [13]; vegetation, water, construction, geometric effects, coherence thresholds and quality masks further structure product support [14]. Deformation can also vary within cities and engineered landscapes [15, 16]. Intersecting population, built-up land or infrastructure with valid radar support defines a support-conditioned domain, appropriate for a target explicitly conditional on observed locations. A broader-domain summary must additionally report allocation outside that support. Missing support alone does not establish deformation at omitted locations.

Coastal and urban studies link subsidence to exposure and flood risk [17, 18], while infrastructure and megacity analyses examine the built environment [19, 20]. Global and regional reviews document its geographic extent and the role of satellite interferometry [21–24]. Nationwide inventories, building-level diagnostics and coastal population-weighted assessments address distinct physical and spatial targets [25–27]. Their population summaries depend on both the observation domain and population allocation; screening workflows must state what remains unresolved outside observed support.

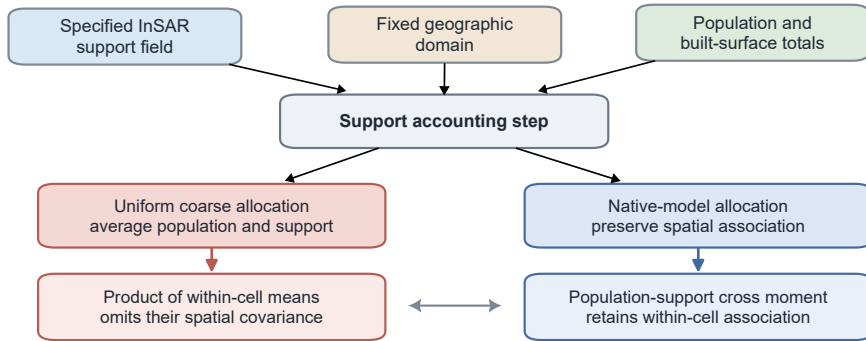
Published assessments also combine quality screening with geodetic comparisons. The Mexico-wide assessment applied temporal-coherence screening and compared InSAR estimates with GPS observations [28], and Ohenhen et al. evaluated rates against 154 GNSS stations across 28 US cities [29]. Published Bangkok studies provide further leveling and CORS comparisons, with their spatial, temporal and station-specific qualifications detailed in Supplementary Section S3 [30–32]. Murray et al. also noted that motion between buildings may remain undetected and require additional observations [33]. These practices and stated limitations provide an essential basis for interpreting displacement measurements and their spatial support.

Population allocation introduces a further step when displacement products are combined with demographic data. Blackwell et al. estimated community exposure assuming uniform population within each community, while Ohenhen et al. assigned census-block population to subsidence classes using each block’s median InSAR-pixel rate [29, 34]. Both studies also used GNSS comparisons. Those comparisons assess displacement agreement; the allocation rules determine how measurements are translated into population summaries. We examine this latter step under a specified observation-support field, holding the domain, support and population total

fixed. These published choices motivate the comparison without implying that either study mishandled missing observations.

Integrating these datasets requires distinguishing the radar support, the physical deformation screen and the population or built-surface layer. Allocation outside the analysis domain differs from allocation whose deformation status is unresolved. Reproducibility requires recording mask application and denominator changes alongside dataset provenance.

Downstream users may combine a released quality mask with a different population grid or reporting geography. Spatial support is part of the data model [35], and aggregation can alter the inferred quantity as recognized by the modifiable areal unit problem [36]. Our matched-input comparison tests whether within-unit population–support association is preserved or replaced by uniform allocation. Areal interpolation and dasymetric mapping provide established approaches to allocating population within coarse units [37, 38]. A finer grid preserves more of the selected model’s spatial structure, without establishing household accuracy. Replacing the average population–support product with the product of separate averages omits within-unit covariance. This established identity explains a possible aggregation effect; it does not determine its empirical size. The questions are whether the effect persists away from boundaries, across population models and support definitions, and when missingness leaves some reporting units without an estimate. We treat public-product observability explicitly, keeping the geographic domain, radar support and population or built-surface allocation separate until accounting. Continuous mean area–pair support replaces the majority threshold as the primary definition. The support deficit measures model allocation unsupported on average across the specified observations; it neither identifies unique unobserved residents nor confirms subsidence without measurements. The earlier binary formulation remains a historical comparator in Section 2.3 and Supplementary Section S2. The audit addresses population-weighted accounting conditional on product support. It does not assess whether displacement measurements are accurate where they exist. Figure 1 summarizes the comparison of uniform coarse allocation and native-model allocation under identical inputs, with threshold, allocation and partition sensitivity reported explicitly [39].



We evaluate population-weighted support accounting under matched inputs. Scale, origin and complete-interior comparisons separate allocation effects from boundary handling. Crossed population-model and pair-selection tests and a same-grid comparison of input support and final quality test dependence on data and support definitions. Controlled missingness evaluates conditional error and unestimated allocation; regional cases test transfer. Together, these comparisons identify when uniform allocation is adequate and when population–support association matters. The reproducibility package records inputs, denominators and executable comparisons [40].

We report small effects, sign reversals and unestimated allocation alongside the primary result. Physical interpretation requires geodetic and geotechnical evidence matched to the locations and quantities being assessed.

2 Materials and methods

2.1 Task formulation and notation

The workflow combines a declared geographic domain, reporting grid, support field and population or built-surface layer. It returns supported and unsupported allocation and their sensitivity to support and allocation choices. This accounting is downstream of displacement retrieval; the separate processing experiment below evaluates one specified LiCSAR/LiCSBAS chain.

We distinguish the geographic domain D (area of interest, AOI), support $s(x) \in [0, 1]$, and nonnegative exposure density $\rho(x)$. The primary Chao Phraya domain contains the 1,807,700 GHSL 3-arc-second pixels whose centers lie in the frozen finite footprint of 18,077 archived VLM cells. It is held fixed without velocity-threshold selection. Because the boundary inherits the finite source-product footprint, it does not represent complete coverage of the geographical delta. The archived VLM GeoTIFF matches the Chao Phraya file in the global-delta dataset accompanying Ohenhen et al. [41, 42]. That study used WabInSAR. Our LiCSAR quality sample is a different processing chain and cannot be interpreted as the validity mask of the WabInSAR product. Moreover, the public GeoTIFF lacks an explicit stored-unit tag establishing the archived factor-of-ten conversion. We therefore retain the inherited strong-deformation selection only as a clearly labeled supplementary sensitivity. Neither that conversion nor a cross-product quality proxy determines the primary geographic population denominator.

For pair j , $M_j(x) = 1$ where unwrapped phase is finite, raster-valid and nonzero and, for the joint screens, byte coherence divided by 255 is at least q . The zero rule is product-specific [43]. We evaluate $q = 0.2, 0.3, 0.4$ and unwrapped validity alone. Integration uses each raster’s actual GDAL transform and intersects masks before averaging; separately averaged masks would not yield their spatial intersection. For GHSL pixel p with area A_p , the reported support is

$$s_p = \frac{1}{NA_p} \sum_{j=1}^N \int_p M_j(x) \, dA. \quad (1)$$

Thus s_p is mean area-pair support at the exposure grid. It is distinct from the area whose individual locations pass a fixed number of pairs, from a binary majority classification at a cell center, and from final inversion validity. We assume each GHSL pixel's population or built-surface total is uniform within that pixel when integrating its intersections. Spherical rectangular areas use longitude and sine-latitude boundaries.

With nonnegative allocated exposure E_p , supported allocation is $S_E = \sum_p E_p s_p$, mean support deficit is $U_E = \sum_p E_p (1 - s_p)$, and total allocation is $T_E = \sum_p E_p$, so $S_E + U_E = T_E$. A pixel assigned 100 people and mean support 0.8 contributes 20 population-allocation units to U_E . These are averages over area and pairs, not 20 identifiable residents. Neither population-valued nor built-surface-valued deficits establish deformation, vulnerability or damage.

The weighted support distribution is

$$F_E(t) = \frac{\sum_p E_p \mathbf{1}(s_p < t)}{T_E}, \quad \int_0^1 F_E(t) dt = \frac{U_E}{T_E}. \quad (2)$$

We report the full support distribution. Its integral is checked using sorted support values, not trapezoidal integration of the plotted 101-point curve.

2.2 Input schema and reproducibility contract

Each run requires stable cell identifiers, geometries and areas; a geographic domain declared before accounting; documented observation support; and population or built-surface quantities on a known spatial support. Any deformation selection additionally requires established units and a stated role. Raster masks, measurement polygons and point observations need distinct support definitions. A point is not repeatedly assigned the full population of its containing cell; an areal rule must first be specified and checked for overlap.

A clipped raster without its removed-support record cannot identify the omitted domain. Likewise, population without geographic support cannot be allocated, and infrastructure counts without asset metadata do not establish engineering risk. These input limits prevent a numerical total from being interpreted beyond its target.

Manifests record source versions, transforms, nodata conventions, established units, dates and checksums. Raster validity, coherence, final quality masks and polygon coverage remain separate fields. Intermediate outputs retain the domain, allocation rule, scale, origin and denominators, with per-pair, per-block or per-station records as applicable (Supplementary Sections S1, S4–S5 and S7). Transfers preserve extensive totals on the common domain.

2.3 Workflow architecture

The binary cell formulation separates deformation-screen membership from whether sampled support passes a majority cutoff (Supplementary Fig. S1). This partition defines a coarse-cell selection: a majority threshold does not identify permanently

unobserved locations, and a selected cell does not locate residents over subsiding ground. We retain the original classification as a historical comparator. Continuous support on the fixed geographic domain defines the primary analysis.

Module 1 registers inputs to the common domain, retaining included area in partial boundary cells. Module 2 intersects validity and coherence masks before integrating support at native population resolution. Module 3 fixes geographic and population denominators independently of allocation; the historical deformation-selected partition remains in Supplementary Fig. S1 and Supplementary Section S2. Module 4 computes supported and unsupported allocation under uniform-area and native-model rules with identical totals. Module 5 tests sensitivity to support definition, sampling, population model and reporting units. Recorded input selections define each run; thresholds are not chosen to maximize the deficit.

The center, area-weighted and ancillary-weighted rules represent distinct choices within established areal-interpolation practice [37, 38]. Supplementary Section S6 states their information requirements and which comparisons are actually performed. Native GHSL allocation is the common model reference for these tests; comparisons with deformation imputation or multi-sensor retrieval would require a different target and additional observations.

Extensive first and cross moments are conservatively transferred to a 50-m UTM integration grid (zone 47N) and repartitioned into 250-, 500-, 1,000- and 2,000-m cells. Each size uses four origins, with zero and half-cell shifts. Partial cells retain only included area; population is not extrapolated outside the domain. A 25-m recomputation checks integration sensitivity (Supplementary Sections S7 and S18). Within cell c , let $P_c = \int_c \rho \, dA$, $H_c = \int_c \rho s \, dA$, $A_c = \int_c dA$ and $Q_c = \int_c s \, dA$. Native-model reference deficit is $P_c - H_c$, while uniform allocation gives $P_c(1 - Q_c/A_c)$. Their difference is

$$U_{\text{uniform}} - U_{\text{native}} = \sum_c \left(H_c - \frac{P_c Q_c}{A_c} \right) = \sum_c A_c \text{Cov}_{A,c}(\rho, s). \quad (3)$$

Preserving H_c retains the population–support association. Center sampling instead uses the fine pixel at each coarse center, with zero support outside the domain. We report signed and absolute discrepancies relative to the native-model deficit and fixed AOI total. A complete-interior comparison excludes boundary-crossing cells, with separate subset denominators and containment checked against the transformed native-pixel boundary (Supplementary Section S7). The native reference is a population model, not household truth.

2.4 Study design and evidence roles

Chao Phraya supplies the primary allocation comparison, a separate Bangkok rectangle the connected processing experiment, and Fresno a published observation footprint. Los Angeles and Davis–Sacramento test additional product and regional transfer. These designs have separate support definitions and validation targets. In a fixed Bangkok rectangle (100.35–100.65° E, 13.65–14.05° N), 437 pairs from 2019–2020 with baselines at most 96 days connect 104 acquisition dates. LiCSBAS2

steps 11–16 use the aligned native inputs and fixed defaults without automatic threshold relaxation [10, 44]. Before pixel-quality inspection, 44 edges are randomly held out while preserving date connectivity; 393 training pairs undergo the same processing. Training time-series differences predict relative LOS displacement at the held-out edges, with training-linear and zero-displacement baselines on common evaluable observations. Referencing, exclusions and exact parameters are recorded in Supplementary S17.

Support strata and quality masks use training inputs only. Held-out dates remain represented in training, so shared dates and atmospheric phase restrict interpretation to internal edge reconstruction, not independent velocity accuracy or forecasting. Separately, legacy 28-pair and matched-stack support are compared with full-stack quality, with GHSL counts transferred by source-area fractions. Coherence contributes to mask construction, and the legacy period differs. This evaluates the specified LiCSAR/LiCSBAS chain, not the WabInSAR raster; matched independent Bangkok GNSS validation remains unavailable.

A fixed Fresno rectangle (119.9–119.7° W, 36.6–36.85° N) tests transfer to the DWR 2026Q1 observation-cell footprint [45, 46]. We clip and union all 38,738 intersecting measurement polygons, then allocate GHSL population once to their nonduplicated intersections. Scale/origin comparisons use UTM zone 11N. The polygon layer supplies coverage, not per-cell velocity or uncertainty; its 2015–2026 observation record also differs from the 2020 population epoch. Artificial masking is restricted to the published footprint.

Population agreement and external transfer designs.

Fresno’s irregular 2020 Census blocks use GHSL allocation normalized to P1 totals and known DWR rates for 1 October 2020–1 October 2021. Random masks and one or four contiguous holes remove 10, 30 or 50% of known-rate cells, with 30 repetitions at -3 , -5 and -10 mm yr $^{-1}$. Native/area prevalence and unit-median rules share masks; undefined allocation, common-computable errors and whole-domain bounds are reported separately (S15).

Five-band spatial holdouts with 0-, 1- and 3-km buffers test training-only total recalibration against ACS, not original population-model training. Shared-date omissions assess dependence in the 224-pair pool without velocity re-inversion (S20). Eighteen training-only quality rules evaluate coverage and residuals on 44 excluded edges; held-out errors do not select thresholds (S21).

Fixed Los Angeles and Davis–Sacramento rectangles test three cutoffs, four scales and four origins using archived ARIA products [47, 48] (S29). Fresno municipal land use supplies strict/broader residential eligibility with unmapped areas retained [49] (S27). Archived NGL GNSS LOS comparisons retain both reference frames and three sampling radii, excluding provider and location-selected local controls from all validation metrics. Temporal-model and source-version limits are retained; this external-product check neither refits daily GNSS nor validates Bangkok (S30).

2.5 Exposure layers and allocation

GHSL supplies consistent public population and built-surface layers across the study domains. Comparing uniform-area and native allocation isolates a spatial assumption under identical inputs; neither represents household-level validation. The resulting quantities are model allocations rather than official demographic or loss estimates.

We compare GHSL R2023A epoch-2020 population at 3 arcsec with WorldPop’s Thailand 2020 unconstrained UN-adjusted population-count product at 3 arcsec [50, 51]. Source counts are transferred by spherical source-area fractions, not by interpolation of density values as if they were counts. Native totals, jointly valid coverage and common-total normalized results are all reported. GHSL 2015/2020 epoch comparisons hold resolution at 30 arcsec. Neither population product is local census truth, and epoch differences combine demographic and model effects.

Crossed aggregation sensitivity.

We cross GHSL and WorldPop 2020 with legacy 28-pair and expanded 224-pair support at $q = 0.3$. All combinations use the same two-model common-valid domain and GHSL total, with 1-km cells at four origins and conserved first and cross moments. Differences are reported relative to both the native deficit and common population total. The 25-m check, all 16 cases and pair-selection details remain in Supplementary Section S18; changing pair number also changes composition. WorldCover 2021 v200 supplies 10-m land classes [52]. Their fractional areas within GHSL pixels inherit the within-pixel uniform-allocation assumption. Tree-class allocation does not identify residences, and land-cover, population and radar epochs differ.

2.6 Robustness and bounded-evidence protocol

The legacy sample contains 28 pairs from 2016–2022. The expanded pool selects eight pairs per year–quarter stratum (224 total) using dates and baseline classes before outcome extraction. Nested ranks yield 28, 56, 112 and 224 pairs; 100 stratified draws without replacement per size assess conditional sampling variation. A pre-extraction rank amendment and both rank plans are archived. Year, quarter, baseline and shared-date omissions test composition and dependence. These comparisons do not estimate independent-observation confidence bounds or uncertainty outside the fixed pool (Supplementary Sections S16 and S20).

Controlled masking uses a separate fixed Bangkok rectangle (100.35–100.75° E, 13.55–13.95° N). Random, spatially smoothed and positive/negative log-density-associated scores remove 10%, 30% or 50% of fine pixels, with a fixed seed and 100 repetitions. Uniform-area, center and built-area-weighted rules share masks and coarse totals; built weights fall back to area where zero. Grids contain 5, 10 or 20 native pixels at two origins, not exact metric cells.

Errors use the known removed allocation U_{ref} . Percentile ranges describe introduced-mask variation conditional on the fixed inputs, not population or InSAR accuracy. Built weighting shares information with GHSL and is not independent validation.

We also construct three synthetic hazard fields: positive density association, negative association and an independent spatial field. Rates of -10 or 0 mm yr^{-1} define known threshold membership. For observed threshold allocation T_{obs} and allocation U in pixels missing under a binary final mask, the identification interval is $[T_{\text{obs}}, T_{\text{obs}} + U]$. Its endpoints assign all missing allocation outside or inside the class. Here U is binary-mask missing allocation, not the continuous mean area-pair deficit. The bound is conditional on the observed rates and population model; it is neither a physical prediction nor a confidence interval.

3 Results

3.1 Lead-case support accounting

Figure 2 compares native-model and uniform allocation under the same support field, geographic domain and allocated totals.

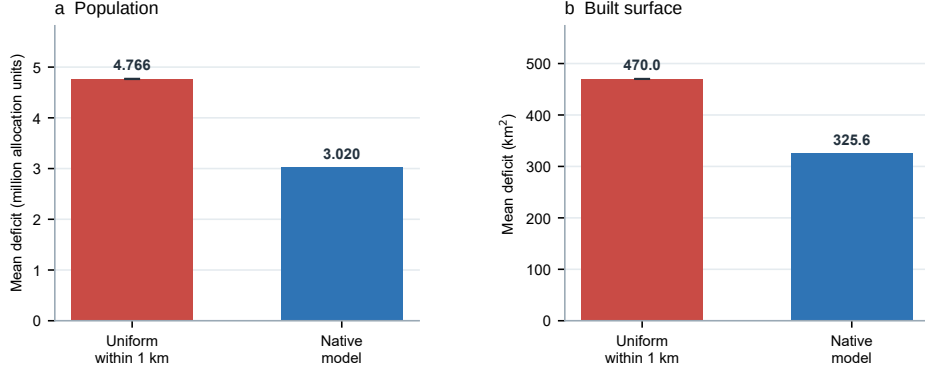
The fixed Chao Phraya footprint contains 22.501 million GHSL population units and 1253.6 km^2 of built surface. At $q = 0.3$, native allocation gives $U_E = 3.020$ million population units (13.42%) and 325.6 km^2 of built surface. Population is concentrated in relatively well-supported portions of the footprint (Supplementary Fig. S14). Consequently, area-, built-surface- and population-weighted support distributions differ (Fig. 3).

Table 1 Native-model support deficits.

Support rule	Population (million units)	Population (%)	Built surface (km^2)	Area (%)
Valid unwrapped	0.268	1.19	31.9	17.89
Joint, $q = 0.2$	1.665	7.40	197.5	57.44
Joint, $q = 0.3$	3.020	13.42	325.6	68.77
Joint, $q = 0.4$	4.891	21.74	464.6	76.83

Unwrapped validity alone and the coherence-screened quantities are substantially different (Table 1). Raising q from 0.2 to 0.4 changes the mean population deficit from 1.665 to 4.891 million. This range is a definition sensitivity, not a confidence interval. The weighted support curves retain information lost by applying one majority threshold to every coarse cell.

Supplementary Section S2 retains the original 3.77-million statistic and input lineage. Its center-majority selection and domain differ from the present mean deficit; the new value is not an estimate of the same quantity or a count of confirmed subsidence exposure.



3.2 Sensitivity to support definition, allocation and population models

With support, native population and AOI total held fixed, uniform allocation within 1-km cells yields 4.763–4.770 million mean population units in deficit, compared with the native-model reference of 3.020 million. This is 57.7–57.9% higher across the four origins. Aggregate allocation discrepancy grows across the examined 250–2,000-m scales (Supplementary Fig. S15). The center approximation also changes with scale and origin; its variability cannot be interpreted as a measurement uncertainty distribution.

The signed discrepancy follows the population–support covariance in Eq. 3. Positive aggregate covariance means uniform redistribution moves population away from better-supported locations, although individual cells can have either sign. Using 25-m instead of 50-m integration changes the 1-km estimate by at most 0.00024 percentage points of AOI population. A separate pixel-center diagnostic verifies the identity and its Cauchy–Schwarz bound (Supplementary Sections S8–S9).

The 1-km effect persists when all boundary-crossing cells are excluded. Across four origins, the complete interior cells contain 99.27–99.49% of domain population, and uniform allocation exceeds their own native-model deficit by 58.24–58.56%. Interior cells account for 98.55–98.87% of the full-domain signed uniform-allocation difference. Thus boundary treatment does not explain the primary allocation effect. Supplementary Section S7 separates complete, partial and outside-center cells and reports both methods under each origin.

On pixels jointly valid in the four population/epoch layers, the $q = 0.3$ deficit is 13.37% for native GHSL 2020 and 23.20% for WorldPop 2020. Joint coverage excludes 142.9 thousand GHSL population units from the full primary domain. Equalizing the two models to the same joint-domain total gives 2.989 versus 5.186 million mean deficit units, so total-population differences do not explain the discrepancy. At fixed 30-arc-second resolution, GHSL’s 2015 and 2020 shares are 20.86% and 20.28%. These epoch sensitivities are smaller than the native-model contrast here, without establishing which model is accurate (Supplementary Fig. S16).

Crossing population model and pair selection.

On the fixed GHSL–WorldPop common domain, both models are normalized to 22.358 million population units. At 1 km, uniform allocation exceeds the native-model deficit by 55.38–55.64% for GHSL with the legacy 28 pairs and 51.29–51.51% with the expanded 224 pairs. The corresponding WorldPop differences are 9.03–9.07% and 8.64–8.68%. All 16 model–sample–origin combinations have positive differences; normalized by the common total, these span 7.40–7.53 percentage points for GHSL and 2.09–2.13 for WorldPop. The direction therefore persists across these inputs, while its magnitude is strongly population-model dependent. The original approximately 58% difference is specific to its full-domain GHSL case. Changing 28 to 224 pairs changes selection and composition as well as sample size; the comparison is not an isolated sample-size effect. Supplementary Section S18 reports every origin and the numerical convergence check.

The Fresno population comparisons separate source totals from spatial allocation. Against 2020 Census blocks, total-normalized 1-km GHSL and WorldPop allocations have weighted absolute percentage error (WAPE; summed absolute unit discrepancies divided by the Census total) of 57–59%, below an area-only baseline (Supplementary Section S24). A 100-m comparison normalizes GHSL and the California population surface (CA-POP) to the same Census block counts; the outside-footprint allocation differs by 0.143 pp, with mixed tract-level directions (Supplementary Section S25). CA-POP uses the same Census counts as inputs, so this is an allocation sensitivity.

Against ACS 2017–2021 estimates for 319 complete Fresno block groups, Census P1, GHSL and WorldPop totals differ by -0.36% , $+10.57\%$ and $+6.03\%$, respectively. Census falls within the aggregate ACS 90% margin of error, without validating household locations. Most block groups are smaller than the 1-km source pixels, so source-cell uniformity remains material. Buffered spatial recalibration gives similar in-sample and held-out errors (Supplementary Sections S20, S26); these reporting-unit checks do not establish fine-scale accuracy.

Municipal residential eligibility leaves about one-third of model allocation unclassified. Its restricted common domain covers 89.13% of complete Census population, below the exploratory 90% reporting gate. Reduced deficits on that domain therefore support only a limited-domain sensitivity (Supplementary Section S27).

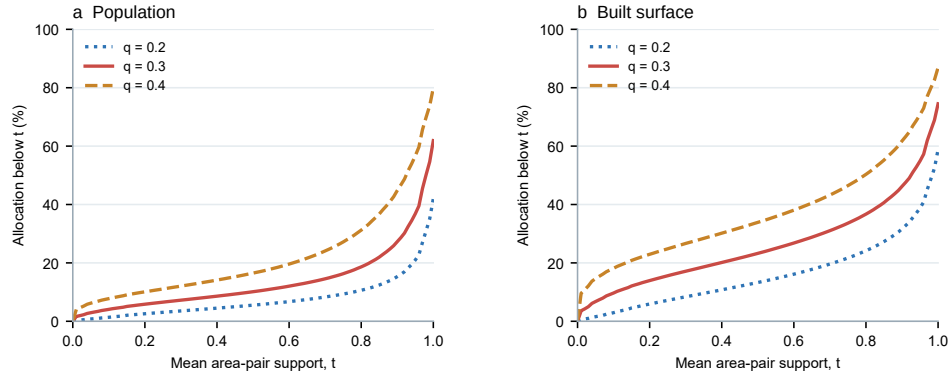
3.3 Sampling uncertainty and its scope

Figure 4 distinguishes conditional variation over draws from the fixed nested selections.

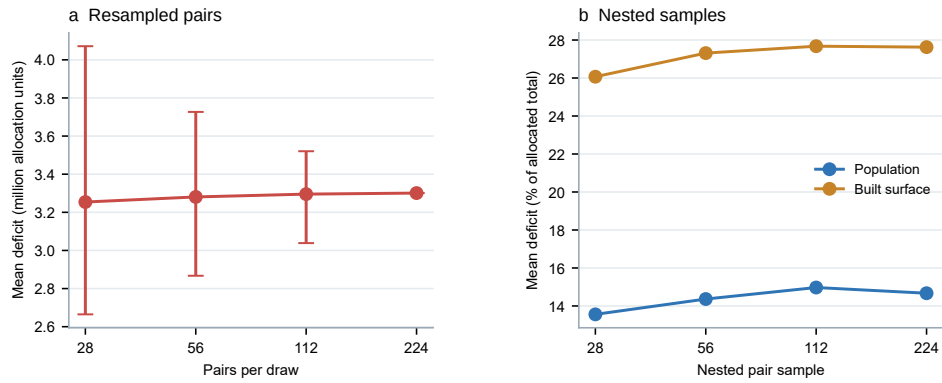
The legacy 28-pair date graph contains 54 dates, 26 components and no independent cycles. It cannot stand in for a complete time-series inversion network. The 224-pair sampling graph is also not fully connected: it has 225 dates and 29 components. In contrast, the 2016–2022 catalogue graph has 288 dates and one component.

Catalogue-level connectivity does not establish pixel-level validity.

The nested 28-, 56-, 112- and 224-pair means give population deficits of 3.051, 3.232, 3.369 and 3.301 million, respectively (Supplementary Fig. S4). The nested 28-pair subset is selected from the expanded pool and differs from the original 28 pairs used



in the primary 3.020-million calculation. Legacy leave-one-year-out results range from 2.820 to 3.181 million. Within the fixed 224-pair pool, the central 95% range of the 100 stratified 28-pair draws is 2.665–4.072 million. That conditional sampling range is wider than the differences between several individual nested means and cannot be converted into a confidence interval for final product quality. Longer-baseline groups have larger mean deficits in both samples. Differences between sample sizes also reflect changes in baseline and seasonal composition. Deleting each shared acquisition date changes the $q = 0.3$ equal-quarter population deficit by -0.386 to $+0.165$ pp. These are dependent support sensitivities, with no velocity re-inversion or confidence interval (Supplementary Section S20).



The checks are summarized by the alternative explanation they address (Table 2). Their different targets preclude treating them as a single validation score.

Table 2 Checks on the accounting result and their interpretation.

Check	Result or comparison	Interpretation
Support definition	Same-grid uniform-minus-native shift: input support 3.066–3.189 pp; final mask 0.108–0.117 pp	Input support is not final-product validity
Scale and grid origin	Complete-interior 1-km difference: 58.24–58.56% of native deficit	Boundary handling does not explain the primary effect
Population and pair selection	Positive shift in all 16 GHSL/WorldPop and 28/224-pair origin cases	Direction persists; magnitude depends on the population model
Controlled missingness	At 30% removal, irregular- block prevalence leaves 0.20% unestimated for random pixels versus 23.01–28.64% for holes	Conditional error and estimator availability must be reported together
Sampling dependence	Fixed-pool repetitions and shared-date omissions	Conditional sensitivity; not independent-observation confidence bounds
Processing and transfer	Excluded pair edges, DWR geometry and external California GNSS	Different evidence targets, summarized in Supplementary Table S29

3.4 Descriptive ranking stability

Spatial rankings are more stable than the totals: across 539 eligible fixed geographic blocks, nested 28-, 56- and 112-pair deficit-amount ranks correlate 0.9940, 0.9992 and 0.9994 with the 224-pair comparator. Top-decile overlaps nevertheless change (Supplementary Fig. S5). This is descriptive stability, not validation of a measurement-priority rule.

Top-decile Jaccard overlap for deficit amount is 0.8000, 0.8621 and 0.9286 for nested 28, 56 and 112 pairs against 224 pairs. Supplementary Fig. S6 reports amount and share rankings. High all-block correlation can coexist with changed short-list membership; monitoring priority would additionally require deformation, consequences and information value.

3.5 Failure-mode experiments and estimator availability

For random and spatial masks unrelated to density, aggregate mean bias is small across repetitions, while the center method has larger sample-to-sample variation. Density-associated masking changes the sign and size of the error (Supplementary Fig. S17). At 30% removal and 10-pixel cells with zero origin, uniform allocation underestimates the known removed population by 14.17% when high-density locations are preferentially removed and overestimates it by 120.35% when low-density locations are preferentially removed. Large relative errors occur when the removed

population is small. We also report errors normalized by the total test-domain population.

Built-area weights reduce error in these GHSL-based scenarios but retain nonzero errors and shared-input dependence. The synthetic hazard experiments additionally show that support deficit alone does not identify threshold-exposed population: the unknown allocation can contain very different hazard shares under the three constructed velocity fields. The conservative identification interval contains the known truth by construction; its width expresses missing information rather than recovered motion. Fresno masking outputs repeat the same protocol and are supplied in the Supplementary Information.

Small conditional errors can coexist with substantial unestimated allocation. At 30% removal, 10-pixel units and -5 mm yr^{-1} , prevalence leaves less than 0.0001% unestimated under random masks but 11.35–28.66% under structured masks. On the common-computable domain, prevalence has smaller local L1 error than median classification for three mechanisms; the median rule is better under low-spread removal. Whole-domain bound widths remain 24.12–34.90 pp (Supplementary Sections S12, S14).

Actual irregular Census blocks show the same availability problem. At 30% removal, prevalence leaves 0.20% unestimated under random masking, versus 23.01% under one contiguous hole and 28.64% under four holes. The median rule leaves 9.41%, 26.80% and 32.61%, respectively. Its whole-block classification also differs from the fine-allocation target before masking. Complete error and bound results are given in Supplementary Section S15.

For the actual LiCSBAS final mask, unsupported allocation is 0.2944% of the fixed domain total. At LOS rate $\leq -5 \text{ mm yr}^{-1}$, observed class allocation is 0.2797% and its whole-domain share is bounded by $[0.2797, 0.5741]\%$, conditional on the observed rates and population model. The 0.2944-pp interval is narrow relative to the domain total but its upper endpoint exceeds twice the observed class share. Both endpoints are attainable by changing only the unknown class assignments. Thus a small unsupported share can still matter for a rare threshold class; no motion is recovered in the gaps (Supplementary Section S13).

3.6 Regional and product transfer

Figure 5 compares the allocation effect and population-weighted coverage under the declared regional support definitions.

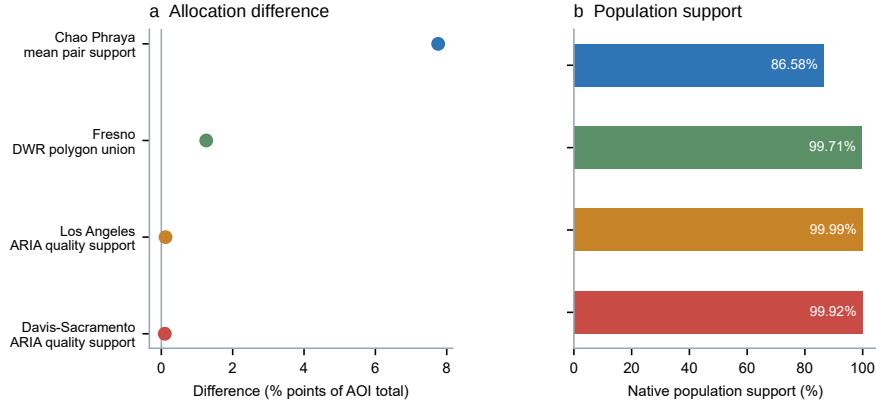
In Fresno, the DWR observation-cell union covers 78.49% of the 495.51-km² rectangle but contains 99.71% of its GHSL allocation. Of 598098 allocation units, 1710 fall outside the footprint. At 1 km, uniform allocation gives 9022–9556 outside-support units, an additional 1.2–1.3 pp of the AOI total. Its large relative multiplier therefore has a small baseline. Center sampling also varies with grid origin (Supplementary Fig. S19).

On the complete Census-block domain, the Census-population-weighted DWR area-coverage share is 99.19%, versus 99.74% for native GHSL allocation (Supplementary Section S23). This normalized-share contrast reflects relative spatial

weights, whereas the 12.20% GHSL–Census total difference is a separate diagnostic. Multiplying all weights by one constant leaves either share unchanged.

A separate DWR rate example uses Census-block median observed vertical-rate classes. The annualized six-window 2015–2021 median covers blocks containing 95.70% of complete-block population, of which 64.99% falls below -5 mm yr^{-1} . Different time-window choices change classification; this transfer does not validate Bangkok LOS rates (Supplementary Section S28).

Across 96 fixed-region scale/origin/cutoff designs, uniform-minus-native differences are only 0.026–0.186 pp in Los Angeles and 0.051–0.372 pp in Davis–Sacramento. The positive direction transfers, but these small effects do not generalize the approximately 58% primary relative difference (Supplementary Section S29).



3.7 Processing checks and external evidence limits

Full-stack quality rejection leaves 431 of 437 pairs for inversion; the training branch retains 382 of 393 inputs. The full quality mask accepts 97.02% of pixels and 99.71% of the small area’s allocated population. Among pixels with matched-input mean support below 0.2, 83.34% still pass the fixed multi-index mask; the 0.8–1 group has 99.72% acceptance. Low coherence-screened support is therefore not equivalent to final processing rejection (Supplementary Fig. S18). Neither acceptance nor rejection is independent physical truth.

Matched-input support bins also differ in median velocity spread (1.343 versus 1.185 mm yr^{-1}) and inversion residual (0.673 versus 0.368 mm, lowest versus highest bins). These are quality associations, not a binary validity surrogate; legacy-sample confusion counts and period differences are reported in Supplementary Section S17. All 44 held-out edges met the prediction and reference requirements. Pooled held-out LOS displacement root-mean-square error (RMSE) is 0.984 mm across evaluable observations. Training-mask accepted and rejected locations give 0.699 and 5.150

mm, respectively; the lowest and highest training-support bins give 2.338 and 0.412 mm. On common evaluable observations, time-series differences, training linear velocity and a zero-displacement baseline give 0.984, 11.751 and 11.784 mm. The held-out pairs connect dates already represented in training. Shared acquisition dates and atmospheric phase limit these residuals to internal reconstruction; they do not measure independent velocity accuracy. Exclusions and per-pair results are retained in Supplementary Section S17.

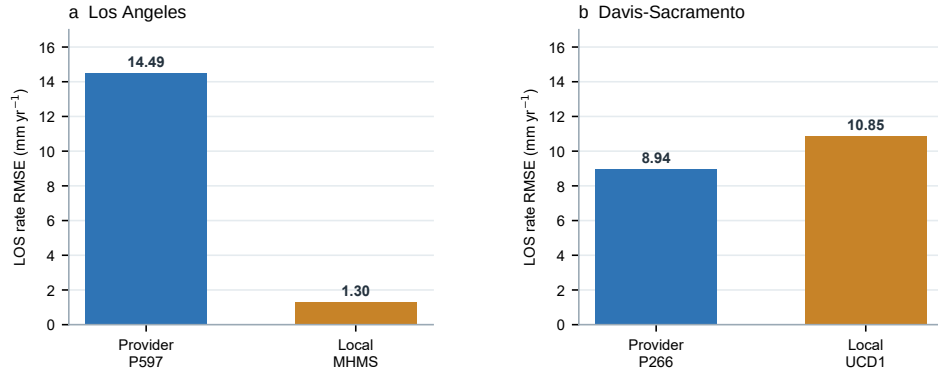
On the same processing grid and GHSL field, uniform allocation within 1-km cells increases the deficit by 3.066–3.189 percentage points of AOI population for mean 437-pair input support, versus 0.108–0.117 points for the final quality mask. Native deficits are 3.4539% and 0.2944%, respectively. Direct source-cell intersections verify the geometry (Supplementary Section S19). This contrast reflects different support definitions, not an error comparison or observed motion in gaps.

Training-only coherence filtering from 0.2 to 0.4 reduces accepted allocation from 99.40% to 95.34% and pooled excluded-edge RMSE from 0.491 to 0.388 mm.

At 0.7 only 41.04% remains. Other metrics have nonmonotonic responses; all 18 fixed-inversion post-filters are reported in Supplementary Section S21, without interpreting residuals as absolute velocity accuracy.

External GNSS agreement is mixed. Native-pixel RMSE in Los Angeles is 14.49 mm yr^{-1} under provider reference P597 and 1.30 under location-selected local control MHMS (11 validation stations). Davis–Sacramento gives 8.94 under P266 and 10.85 under UCD1 (four stations). Controls are excluded from validation; every sampling radius and both frames are retained. Archived source-version conflicts, temporal-model differences and station dependence limit interpretation. These results cannot be transferred to our Bangkok inversion (Supplementary Section S30).

Figure 6 retains both reference-frame choices in each region.



Supplementary Fig. S20 and Supplementary Table S29 distinguish accounting checks, internal reconstruction and external-product comparisons from the unresolved physical validation of the Bangkok inversion.

4 Discussion

4.1 Earth-science informatics implication

The central implication is that public InSAR observability should be treated as part of the information product when deformation maps are translated into exposure estimates. A support mask specifies the observation domain and support available to a downstream analysis. The proposed audit links that support to a traceable population or built-surface allocation quantity.

The matched-input comparisons establish a material change in population-weighted support summaries when fine spatial allocation is replaced by uniform coarse allocation. The effect persists in complete interior cells, so boundary handling does not explain the primary result. Crossing population models and pair selections gives a consistent positive direction across the 16 tested cases, but the relative difference is much smaller with WorldPop than with GHSL. The approximately 58% difference is specific to the primary GHSL case; the crossed comparisons show how the magnitude changes with the population distribution. It does not establish a universal correction factor.

Measurement precision and coherent-scatterer reliability do not fully describe uncertainty in exposure accounting. A product can be accurate where it observes while a population summary remains conditional on that support. Reporting the continuous support deficit makes explicit the mean population allocation unsupported across the specified interferogram pairs. This complements the quality information supplied by the radar processor.

For city analysts, infrastructure planners and hazard-screening researchers, the support mask and population-allocation rule must remain explicit when a public deformation layer is combined with demographic data. Supported and unsupported totals provide a shared interpretation for radar specialists and downstream users who do not revisit the full processing chain.

Reusable products therefore need machine-readable support semantics as well as values. Velocity, coherence, validity and population belong to different data layers. Keeping them separate distinguishes zero exposure from unsupported allocation and allocation outside the domain; the final table should preserve those meanings. Related support-and-allocation questions arise when other Earth-observation products are combined with human-exposure layers or administrative boundaries, including flood, fire and coastal-impact screening. Such applications require their own physical targets and tests; the present experiments establish results only for the stated InSAR and population inputs.

4.2 Relationship to radar and signal-processing practice

The support deficit extends measurement-support and missingness analysis into population-weighted accounting. A processing chain retains coherent or otherwise valid measurements unevenly across space and time; its product therefore need not represent the geographic domain uniformly.

Water, vegetation, surface change and construction make missing support geographically structured. Urban deltas and coastal plains combine dense development, waterways, soft soils and construction zones. Subsidence and exposure intersect in coastal Semarang [53]; time-series radar measures coastal urban subsidence in Jakarta [54]. Wuhan and Karapinar extend the examples to inland urban and basin settings [55, 56]. Where observability correlates with population, the support mask changes the population summary.

The covariance relationship explains the primary overestimation and the opposite errors under density-associated masking. Small aggregate biases under density-independent masks identify conditions in which uniform allocation can be adequate. Preserving finer model structure is useful only insofar as that structure is appropriate to retain; agreement with a native-model reference does not establish household accuracy.

Measurement support and measurement value remain distinct. Low support does not itself indicate subsidence. The primary geographic domain is therefore fixed without a velocity threshold, while the support field records mean area-pair availability. A physical deformation screen would require independently interpretable rate measurements.

Reconstructing deformation in missing locations would require additional observations or assumptions about geology, groundwater, land cover, loading and spatial correlation. Those assumptions may be justified locally but cannot be inferred from a support deficit. Here, numerical correctness is assessed conditionally on specified inputs without asserting deformation in gaps.

Ascending/descending geometries, local GNSS, leveling, piezometry or building observations could support a separate validation or attribution analysis. A verified deformation field would then permit population accounting within physical threshold classes, with unresolved locations still reported. Choosing new measurement sites would additionally require information value and local consequences.

4.3 Use for infrastructure and power-system screening

The workflow could use substations, transmission-line length, distribution corridors or other critical assets in place of E_p . An asset-specific support definition would describe incomplete radar coverage, while risk assessment would also require design, age, redundancy, foundation and maintenance information. The present experiments evaluate neither these assets nor their failure probabilities or power-system risk. Combining support at asset locations or corridors with engineering observations, asset criticality and information value could inform follow-up studies. Vulnerability and intervention priorities require engineering assessment.

4.4 When the audit should and should not be used

The audit is useful when reusing a deformation product with retained support information and allocatable exposure layers, particularly when restricting to visible support changes the denominator. Before local validation, it describes where

the selected population model has weaker support. It cannot determine where measurement or intervention would deliver the greatest benefit.

For dense site-level diagnosis, matched leveling, GNSS, borehole, hydrogeological and engineering observations may make public-product support a secondary concern. The audit can document support but cannot replace that evidence. If a product has discarded invalid areas without retaining its mask, omitted support cannot be reconstructed reliably; the appropriate result is a data-limitation statement, not an unsupported-allocation estimate for that domain.

A complete-case overlay describes allocation within observed support. Center sampling is inexpensive but can miss subcell structure and boundary coverage. Uniform-area transfer is suitable when its approximation error is acceptable for the intended summary. Building weights add assumptions and may share population-model inputs. Imputation, multiple sensors and ground measurements add deformation information and require separate evaluation; our experiments assess their accompanying support-accounting choices.

4.5 Limitations

The strongest physical limitation is the absence of matched independent GNSS or leveling validation for the Bangkok inversion. A local feasibility audit found that CUSV and CUUT occupy the same InSAR pixel and lack an independently established datum shared with the inversion (Supplementary Section S17). Controlled masks test accounting against conditional references; excluded pair edges test internal reconstruction. California GNSS supplies independent observations for a different product and different regions, with mixed results under two reference-frame choices. None identifies deformation in permanently unsupported Bangkok locations (Supplementary Fig. S20; Supplementary Table S29).

This limitation restricts physical interpretation. A support deficit cannot be converted into confirmed affected population without evidence about deformation at the relevant locations. The controlled experiments instead establish what changes when the same inputs are combined under different allocation rules. The primary Chao Phraya domain inherits the finite source-VLM footprint, so its conclusions do not establish coverage of the whole delta or extend to locations outside that footprint. The additional rectangles likewise do not establish a worldwide magnitude of the effect.

The original majority-observable partition simplifies a continuous support problem. The revised primary analysis therefore retains support fractions and the full population-weighted support distribution. This change addresses threshold information loss, but it does not make coherence-screened pair support equivalent to reliable velocity or to a final processing mask. Sampling variation remains conditional on the archived pair pool; shared acquisitions and atmospheric signals also limit the interpretation of held-out edge residuals. Pair number, seasonal composition and temporal baseline are not isolated experimental factors.

GHSL provides reproducible public inputs but cannot replace local census, cadastral or infrastructure inventories. Population grids may misallocate residents in changing urban areas; built-up products may represent informal settlement, industrial

surfaces and mixed land cover differently. WorldPop, Census, CA-POP and ACS comparisons measure disagreements at stated supports without fully propagating population-product uncertainty or establishing household locations. CA-POP shares Census inputs, and land-cover or built-area weights may share information with GHSL. Incomplete municipal residential classification restricts the common domain, so its results are not population-wide validation.

The workflow supports screening, not causal attribution. Subsidence can arise from groundwater withdrawal, sediment compaction, loading, hydrocarbon extraction, tectonics or other processes. National assessments show variation across hydrogeological and urban settings [57]; coastal policy studies show that control benefits and priorities depend on local exposure and management [58]. Attribution requires hydrogeological, geological and geotechnical evidence beyond these inputs. Further physical interpretation also requires independent deformation observations matched to the lead domain and stronger local population or asset data.

Earlier estimates and source attribution.

The submitted 3.772492-million center-majority total is reproducible, but it is not a verified count of people exposed to subsidence in missing areas. The earlier center-majority statistic differs from the mean support deficit defined here; their relationship and input provenance are documented in Supplementary Section S2. We have removed the exploratory Iran branch from quantitative applications. Its archived rate and mask correspond to the 2014–2020 product associated with Haghshenas Haghighi and Motagh (2024). No Iran result or new/old Iranian product comparison contributes to the quantitative conclusions. Supplementary Section S3 documents the attribution correction and distinguishes the newer products of Payne et al. [12, 59, 60].

5 Conclusions

Spatial aggregation changes population-weighted InSAR support summaries by omitting the within-cell population–support cross moment. Fixed-input and complete-interior comparisons establish this effect. Its direction persists across the tested population models and pair selections, while its magnitude depends on the inputs and study area. Controlled missingness identifies conditions under which the error reverses, aggregate bias is small or substantial allocation remains unestimated. The published-support and regional comparisons show that transfer also depends on the support definition; external GNSS agreement differs between regions. Population-weighted reporting should therefore specify its domain and support field, preserve extensive quantities and their cross moments, and report sensitivity to allocation and sampling. Deformation in unsupported locations remains unknown.

Screening with this workflow requires a declared domain, population model, support field and allocation rule; summaries restricted to observed support must be labelled accordingly. Site diagnosis and intervention still require denser geodetic, demographic and geotechnical evidence.

Data Availability

Public input data are obtained from COMET LiCSAR, GHSL, WorldPop, ESA WorldCover, the U.S. Census Bureau, California DWR, the archived ARIA California products, NGL and City of Fresno municipal GIS. Exact source-product identifiers, acquisition methods and input checksums are documented in the Supplementary Information and accompanying reproducibility materials. Source-provider terms govern access to and reuse of third-party inputs. Frozen processed analysis inputs, full reference and replicate tables, figure source data and the manuscript number ledger are publicly available with the versioned scientific reproduction package.

Code Availability

The scientific reproduction package is publicly available at the fixed GitHub release [v0.6.2](#). It contains the experiment code, configurations, input manifests and checksums, processed analysis inputs, full reference outputs, environment records and reviewer reproduction instructions. Companion assets supply portable inputs and selected complete source products for S15, S20–S21, S27, S29–S30, with file-level checksums. The complete ACS files and official Census source archives used in S25 remain available at v0.4.1. The reviewer guide provides reproduction and numerical-comparison commands for the processed inputs and reference outputs retained from release v0.6.0. The current manuscripts, two portable local audits and a section-to-output index are identified separately under `revision_2026/submission_20261003/`; these current documents supersede earlier manuscript snapshots in the archive. Other provider-controlled original products and the complete interferogram processing stacks are obtained separately through the documented acquisition routes. The earlier DOI [10.5281/zenodo.21444768](#) archives v0.3.0; a revised DOI deposit remains pending.

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Author Contributions

Zixuan Liu: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing—original draft, Writing—review and editing. Wei Xiong: Conceptualization, Supervision, Methodology, Validation, Writing—review and editing.

Additional Information

Competing interests

The authors declare no competing interests.

Figure legends

Figure 1. Workflow for population-weighted support accounting. The comparison fixes the geographic domain, support field and allocated total. It compares uniform coarse allocation with preservation of the native population–support cross moment. Neither branch infers motion in unsupported locations.

Figure 2. Chao Phraya population and built-surface support deficits in the fixed geographic domain at $q = 0.3$. Red bars show uniform allocation within 1-km cells averaged across four origins; narrow whiskers span those origins. Blue bars show the native-model reference. These are conditional model-allocation quantities, not unique residents or confirmed subsidence exposure. The submitted 3.77-million statistic uses a different domain and binary selection; its lineage is preserved in Supplementary Section S2.

Figure 3. Continuous support sensitivity. Curves give the fraction of allocated population or built-surface quantity with native-pixel mean support below t , for joint unwrapped/coherence cutoffs $q = 0.2, 0.3, 0.4$. The domain and allocated totals are identical across cutoffs. The display uses 101 threshold values; the exact empirical-distribution integral is checked separately. No velocity threshold determines these curves.

Figure 4. Pair-sampling sensitivity. (a) Means and central 95% ranges from 100 draws at each size within the fixed 224-pair pool, at $q = 0.3$. At 224 pairs the whole pool is selected and the range collapses. (b) Population and built-surface deficit shares in the fixed nested selections. The nested 28 pairs differ from the legacy 28-pair sample. Ranges condition on the archived pool; neither panel supplies calibrated uncertainty for household population or true deformation.

Figure 5. Regional transfer under specified support fields. (a) Uniform-minus-native population support deficits for 1-km cells, in percentage points of each fixed AOI total; whiskers span four grid origins. (b) Native population-weighted support in the same domains. Chao Phraya uses sampled mean area–pair support at $q = 0.3$; Fresno uses the nonduplicated

DWR observation-polygon union; Los Angeles and Davis–Sacramento use their documented ARIA quality-support definitions at the displayed 0.3 cutoff. The support objects differ, and the effects are not pooled or interpreted as a universal correction factor.

Figure 6. External GNSS consistency checks. Native-pixel California InSAR–GNSS LOS-rate RMSE is shown for the archived provider frame and the location-selected local frame. Control sites are excluded from the 11 Los Angeles and four Davis–Sacramento validation stations. Both reference choices are retained. Temporal-model, source-version and station-dependence limitations are detailed in Supplementary Section S30. These comparisons do not validate the separate Bangkok inversion or genuinely missing deformation.

Table legends

Table 1. Native-model support deficits in the primary geographic domain and legacy 28-pair sample. Each percentage uses its stated population or area denominator.

Table 2. Checks on support accounting, grouped by the alternative explanation or information limitation they address. The table summarizes experiments detailed in the main Results and Supplementary Sections S4–S30; it is not a combined validation score.